

It's a mouthful, but it's a good workable definition. In the following sections, we'll add some meat to this rather bare-bones definition of network management.

9.2 The Infrastructure for Network Management

We've seen in the preceding section that network management requires the ability to “monitor, test, poll, configure, . . . and control” the hardware and software components in a network. Because the network devices are distributed, this will, at a minimum, require that the network administrator be able to gather data (for example, for monitoring purposes) from a remote entity and effect changes at that remote entity (for example, control it). A human analogy will prove useful here for understanding the infrastructure needed for network management.

Imagine that you're the head of a large organization that has branch offices around the world. It's your job to make sure that the pieces of your organization are operating smoothly. How will you do so? At a minimum, you'll periodically gather data from your branch offices in the form of reports and various quantitative measures of activity, productivity, and budget. You'll occasionally (but not always) be explicitly notified when there's a problem in one of the branch offices; the branch manager who wants to climb the corporate ladder (perhaps to get your job) may send you unsolicited reports indicating how smoothly things are running at his or her branch. You'll sift through the reports you receive, hoping to find smooth operations everywhere but no doubt finding problems in need of your attention. You might initiate a one-on-one dialogue with one of your problem branch offices, gather more data in order to understand the problem, and then pass down an executive order (“Make this change!”) to the branch office manager.

Implicit in this very common human scenario is an infrastructure for controlling the organization—the boss (you), the remote sites being controlled (the branch offices), your remote agents (the branch office managers), communication protocols (for transmitting standard reports and data, and for one-on-one dialogues), and data (the report contents and the quantitative measures of activity, productivity, and budget). Each of these components in human organizational management has a counterpart in network management.

The architecture of a network management system is conceptually identical to this simple human organizational analogy. The network management field has its own specific terminology for the various components of a network management architecture, and so we adopt that terminology here. As shown in Figure 9.2, there are three principal components of a network management architecture: a managing entity (the boss in our analogy above—you), the managed devices (the branch office), and a network management protocol.

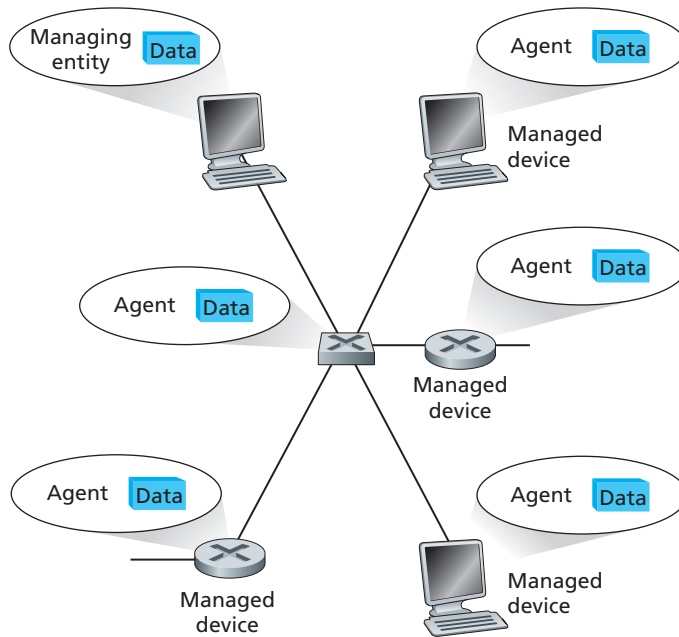


Figure 9.2 ♦ Principal components of a network management architecture

The **managing entity** is an application, typically with a human in the loop, running in a centralized network management station in the NOC. The managing entity is the locus of activity for network management; it controls the collection, processing, analysis, and/or display of network management information. It is here that actions are initiated to control network behavior and here that the human network administrator interacts with the network devices.

A **managed device** is a piece of network equipment (including its software) that resides on a managed network. This is the branch office in our human analogy. A managed device might be a host, router, bridge, hub, printer, or modem. Within a managed device, there may be several so-called **managed objects**. These managed objects are the actual pieces of hardware within the managed device (for example, a network interface card), and the sets of configuration parameters for the pieces of hardware and software (for example, an intradomain routing protocol such as RIP). In our human analogy, the managed objects might be the departments within the branch office. These managed objects have pieces of information associated with them that are collected into a **Management Information Base**

(MIB); we'll see that the values of these pieces of information are available to (and in many cases able to be set by) the managing entity. In our human analogy, the MIB corresponds to quantitative data (measures of activity, productivity, and budget, with the latter being settable by the managing entity!) exchanged between the branch office and the main office. We'll study MIBs in detail in Section 9.3. Finally, also resident in each managed device is a **network management agent**, a process running in the managed device that communicates with the managing entity, taking local actions at the managed device under the command and control of the managing entity. The network management agent is the branch manager in our human analogy.

The third piece of a network management architecture is the **network management protocol**. The protocol runs between the managing entity and the managed devices, allowing the managing entity to query the status of managed devices and indirectly take actions at these devices via its agents. Agents can use the network management protocol to inform the managing entity of exceptional events (for example, component failures or violation of performance thresholds). It's important to note that the network management protocol does not itself manage the network. Instead, it provides capabilities that a network administrator can use to manage ("monitor, test, poll, configure, analyze, evaluate, and control") the network. This is a subtle, but important, distinction.

Although the infrastructure for network management is conceptually simple, one can often get bogged down with the network-management-speak vocabulary of "managing entity," "managed device," "managing agent," and "Management Information Base." For example, in network-management-speak, in our simple host-monitoring scenario, "managing agents" located at "managed devices" are periodically queried by the "managing entity"—a simple idea, but a linguistic mouthful! With any luck, keeping in mind the human organizational analogy and its obvious parallels with network management will be of help as we continue through this chapter.

Our discussion of network management architecture above has been generic, and broadly applies to a number of the network management standards and efforts that have been proposed over the years. Network management standards began maturing in the late 1980s, with OSI **CMISE/CMIP** (the **Common Management Information Services Element/Common Management Information Protocol**) [Piscatello 1993; Stallings 1993; Glitho 1998] and the Internet **SNMP** (**Simple Network Management Protocol**) [RFC 3410; Stallings 1999; Rose 1996] emerging as the two most important standards [Subramanian 2000]. Both are designed to be independent of vendor-specific products or networks. Because SNMP was quickly designed and deployed at a time when the need for network management was becoming painfully clear, SNMP found widespread use and acceptance. Today, SNMP has emerged as the most widely used and deployed network management framework. We'll cover SNMP in detail in the following section.



PRINCIPLES IN PRACTICE

COMCAST'S NETWORK OPERATIONS CENTER

Comcast's world-class fiber-based IP network delivers converged products and services to 49 million combined video, data and voice customers. Comcast's network includes more than 618,000 plant route miles, 138,000 fiber route miles, 30,000 backbone miles, 122,000 optical nodes, and massive storage for the Comcast Content Delivery Network, which delivers a Video on Demand product of more than 134 Terabytes. Each part of Comcast's network, up to and including the customers' homes or businesses, is monitored by one of the company's Operations Centers.

Comcast operates two National Network Operations Centers that manage the national backbone, regional area networks, national applications and specific platforms supporting voice, data and video infrastructure for residential, commercial and wholesale customers. In addition, Comcast has three Divisional Operations Centers that manage the local infrastructure that supports all of their customers. Both the National and Divisional Operations Centers are accountable for proactively monitoring all aspects of their network and product performance on a 7 x 24 x 365 basis, utilizing common processes and systems. For example, various network events at the national and local levels have common pre-defined severity levels, recovery processes, and expected Mean Time to Restore objectives. The national and divisional centers can back up each other if a local issue impacts a site's operation. In addition, the National and Divisional Operations Centers have an extensive Virtual Private Network that allows engineers to securely access the network to remotely perform proactive or reactive network management activities.

Comcast's approach to network management involves five key areas: Performance Management, Fault Management, Configuration Management, Accounting Management and Security Management. **Performance Management** is focused on understanding



These screens show tools supporting correlation, threshold management, ticketing used by Comcast technicians (Courtesy of Comcast.)

(continues)

how the network/systems and applications (collectively referred to as the ecosystem) are performing with respect to pre-defined measures specific to time of day, day of week, or special events (e.g., storm surges or pay events, such as a boxing match). These pre-defined performance measures exist throughout the service path, from the customer's residence or business through the entire network, as well as the interface points to partners and peers. In addition, synthetic transactions are run to ensure the health of the ecosystem on a continual basis. **Fault Management** is defined as the ability to detect, log and understand anomalies that may impact customers. Comcast utilizes correlation engines to properly determine an event's severity and act appropriately, eliminating or remediating potential issues before they affect customers. **Configuration Management** makes sure appropriate versions of hardware and software are in place across all elements of the ecosystem. Keeping these elements at their peak "golden" levels helps them avoid unintended consequences. **Accounting Management** ensures that the operations centers have a clear understanding of the provisioning and utilization of the ecosystem. This is especially important to ensure that at all times the operations centers have the ability to re-route traffic effectively. **Security Management** ensures that the proper controls exist to ensure the ecosystem is effectively protected against inappropriate access.

Network Operations Centers and the ecosystem they support are not static. Engineering and Operations personnel are constantly re-evaluating the pre-defined performance measures and tools to ensure that the customers' expectations for operational excellence are met.

9.3 The Internet-Standard Management Framework

Contrary to what the name SNMP (Simple Network Management Protocol) might suggest, network management in the Internet is much more than just a protocol for moving management data between a management entity and its agents, and has grown to be much more complex than the word "simple" might suggest. The current Internet-Standard Management Framework traces its roots back to the Simple Gateway Monitoring Protocol, SGMP [RFC 1028]. SGMP was designed by a group of university network researchers, users, and managers, whose experience with SGMP allowed them to design, implement, and deploy SNMP in just a few months [Lynch 1993]—a far cry from today's rather drawn-out standardization process. Since then, SNMP has evolved from SNMPv1 through SNMPv2 to the most recent version, SNMPv3 [RFC 3410], released in April 1999 and updated in December 2002.

When describing any framework for network management, certain questions must inevitably be addressed: